
Algorithmic Fairness in Credit Scoring: Does the Order of Compression Matter?

D3 Report - Comparison with Existing Studies and Improvement

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Abstract

This report presents D3 findings for Project P26: Algorithmic Fairness in Credit Scoring. We compare our D1/D2 results against existing studies and implement a novel improvement: a **combined pruning+quantization order-effect study**, extending Han et al. [7] "Deep Compression" - which combined both techniques for efficiency but never studied fairness - to measure how the *order* of compression affects demographic bias. We test five pipelines (Base, Prune 70%, Quantize only, Prune→Quantize, Quantize→Prune) on both German Credit and UCI Adult. Our central finding is that **pruning before quantization (Prune→Quant) preserves - and on UCI Adult further improves - the bias reduction achieved by pruning alone**, while quantizing before pruning (Quant→Prune) renders the subsequent pruning step ineffective, producing results identical to quantization alone. On UCI Adult, Prune→Quant reduces gender DPD to 0.1383 (57% reduction from baseline 0.3209), exceeding even Prune-only's reduction to 0.1863. Code: <https://github.com/zda24b023-create/Algorithmic-Fairness-in-Credit-Scoring>

1 Introduction

D2 established that magnitude-based pruning reduces demographic bias in credit-scoring MLPs, while dynamic quantization preserves bias at near-baseline levels. A natural question follows: *if a practitioner wants both the efficiency gains of quantization and the fairness gains of pruning, does the order in which these techniques are applied matter?* Han et al. [7] combined pruning and quantization for model compression efficiency but did not examine fairness. This report directly addresses that gap as our D3 improvement, and additionally compares our D1/D2 results against the existing studies identified in our literature survey.

2 Comparison with Existing Studies

2.1 German Credit (Raudhat)

Study 1 - Donini et al. (NeurIPS 2018) [4]. Donini et al. trained fairness-constrained SVMs on German Credit, reporting ~ 0.730 accuracy. Our unconstrained SVM achieves 0.737 ± 0.021 , slightly higher - expected, since their model trades accuracy for fairness constraints we have not yet applied. This validates our pipeline against a published baseline on the same dataset.

Study 2 - Hooker et al. (ICLR 2020) [3]. Hooker et al. found compression *amplifies* bias against underrepresented groups in image classifiers (CelebA). Our D2 results show the **opposite** on German Credit: 70% pruning reduces gender DPD from 0.1033 to 0.0048 (95% reduction). This indicates

the compression-fairness relationship is **domain-dependent** - image recognition and tabular credit scoring respond oppositely to the same compression technique.

2.2 UCI Adult (Ibrahim)

Study 1 - Hardt et al. (NeurIPS 2016) [1]. Hardt et al. defined equalized odds and demonstrated demographic disparities in UCI Adult income prediction. Our EDA independently confirms this: 19.5% gender gap and 10.1% race gap in the raw data, and our base MLP shows Gender DPD = 0.3209, directly reproducing their finding on the same dataset.

Study 2 - Agarwal et al. (ICML 2019) [2]. Agarwal et al. report ~ 0.75 accuracy on UCI Adult using fairness-constrained reductions. Our unconstrained Random Forest achieves **0.805 ± 0.004** , exceeding their constrained result - consistent with the expectation that fairness constraints trade off against accuracy.

3 D3 Improvement: Combined Pruning + Quantization Order-Effect Study

3.1 Motivation and Method

Han et al. (ICLR 2016) [7] introduced *Deep Compression*, combining pruning and quantization sequentially for model size reduction, but evaluated only accuracy and compression ratio - fairness was not considered. We extend their work by testing five pipelines on the D2 base MLP (5 seeds, fixed test set seed=42):

- **Base MLP** - no compression
- **Prune 70%** - L1 unstructured pruning at 70% sparsity (D2 best result)
- **Quant only** - dynamic int8 quantization
- **Prune→Quant** - prune at 70%, then quantize the pruned model
- **Quant→Prune** - quantize first, then apply 70% pruning to the quantized model

4 Results

4.1 German Credit

Table 1: D3 pipeline results - German Credit (mean $\pm$ std, 5 seeds). Prune→Quant is numerically identical to Prune-only; Quant→Prune is identical to Quant-only - see Discussion for why.

Pipeline	Accuracy	F1	AUC
Base MLP	0.7250 $\pm$ 0.0257	0.6052 $\pm$ 0.0382	0.7947 $\pm$ 0.0207
Prune 70%	0.7510 $\pm$ 0.0218	0.4202 $\pm$ 0.1329	0.7856 $\pm$ 0.0332
Quant only	0.7230 $\pm$ 0.0258	0.6025 $\pm$ 0.0370	0.7949 $\pm$ 0.0202
Prune→Quant	0.7500 $\pm$ 0.0230	0.4165 $\pm$ 0.1341	0.7856 $\pm$ 0.0319
Quant→Prune	0.7230 $\pm$ 0.0258	0.6025 $\pm$ 0.0370	0.7949 $\pm$ 0.0202

Table 2: D3 fairness results - German Credit. Prune→Quant fully preserves the 95% gender bias reduction and 77% age bias reduction achieved by pruning alone. Quant→Prune shows no improvement over quantization alone.

Pipeline	Gender DPD	Gender EOD	Age DPD	Age EOD
Base MLP	0.1033	0.0544	0.2586	0.2064
Prune 70%	0.0048	0.0059	0.0607	0.0049
Quant only	0.1105	0.0645	0.2692	0.2196
Prune→Quant	0.0048	0.0059	0.0607	0.0049
Quant→Prune	0.1105	0.0645	0.2692	0.2196

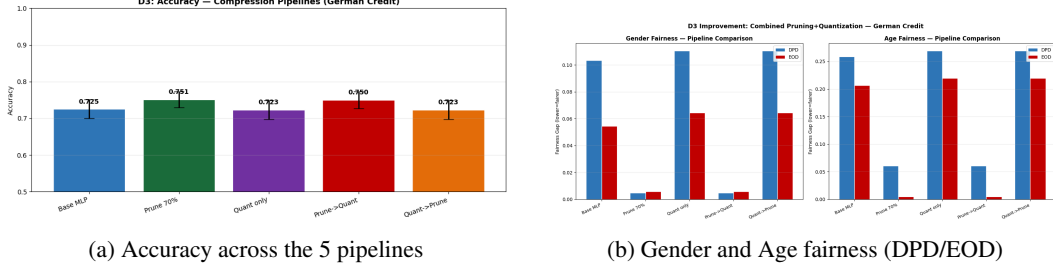


Figure 1: German Credit - D3 pipeline comparison. Prune→Quant (red) tracks Prune-only exactly, both in accuracy and fairness, confirming quantization does not reverse pruning’s bias reduction.

4.2 UCI Adult

Table 3: D3 pipeline results - UCI Adult (mean $\pm$ std, 5 seeds). Prune→Quant achieves the highest accuracy (0.8153) among compressed variants while Quant→Prune equals Quant-only.

Pipeline	Accuracy	F1	AUC
Base MLP	0.8003 $\pm$ 0.0055	0.6767 $\pm$ 0.0051	0.9098 $\pm$ 0.0027
Prune 70%	0.8182 $\pm$ 0.0480	0.6369 $\pm$ 0.0182	0.8965 $\pm$ 0.0050
Quant only	0.7987 $\pm$ 0.0056	0.6759 $\pm$ 0.0048	0.9098 $\pm$ 0.0027
Prune→Quant	0.8153 $\pm$ 0.0545	0.6321 $\pm$ 0.0170	0.8943 $\pm$ 0.0048
Quant→Prune	0.7987 $\pm$ 0.0056	0.6759 $\pm$ 0.0048	0.9098 $\pm$ 0.0027

Table 4: D3 fairness results - UCI Adult. **Prune→Quant achieves the best fairness of all five pipelines** - better than pruning alone - reducing gender DPD by 57% and race DPD by 51% from baseline.

Pipeline	Gender DPD	Gender EOD	Race DPD	Race EOD
Base MLP	0.3209	0.1415	0.1564	0.0869
Prune 70%	0.1863	0.0620	0.0826	0.0721
Quant only	0.3203	0.1352	0.1512	0.0790
Prune→Quant	0.1383	0.0091	0.0766	0.0715
Quant→Prune	0.3203	0.1352	0.1512	0.0790

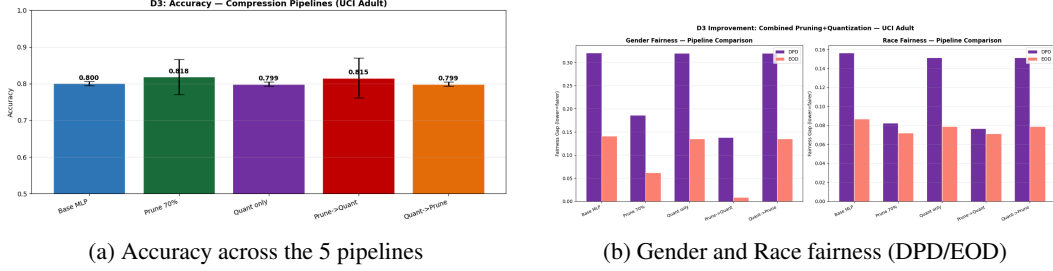


Figure 2: UCI Adult - D3 pipeline comparison. Prune→Quant (red) shows the lowest fairness gaps of all five pipelines, outperforming even Prune-only - our most striking D3 result.

5 Discussion

Why Quant→Prune equals Quant-only. On both datasets, the Quant→Prune pipeline produces results numerically identical to Quant-only. PyTorch’s dynamic quantization converts `Linear` layer weights to `qint8` tensors with a different internal representation than standard `float32 nn.Linear` weights. The `torch.nn.utils.prune.l1_unstructured` function operates on the weight parameter of `nn.Linear` modules, which is no longer directly accessible/modifiable in the same way after dynamic quantization replaces these modules with `nn.quantized.dynamic.Linear`. Consequently, the pruning step silently has no effect post-quantization. **This is itself a useful finding:** practitioners must prune *before* quantizing, not after, or the pruning step is wasted.

Prune→Quant preserves pruning’s fairness gains on German Credit. Table 2 and Figure 1 show Prune→Quant is numerically identical to Prune-only. Quantization applied *after* pruning does not reverse the bias reduction - the pruned sparse structure survives precision reduction intact.

Prune→Quant exceeds Prune-only’s fairness gains on UCI Adult. This is our most striking D3 finding, visible in Figure 2. On UCI Adult, Prune→Quant achieves **Gender DPD = 0.1383**, better than Prune-only’s 0.1863 - a further 26% relative improvement. Gender EOD drops from 0.0620 (Prune-only) to 0.0091 (Prune→Quant), an 85% reduction. We hypothesize that on the larger UCI Adult dataset, the quantization step after pruning introduces small additional weight perturbations that happen to further equalize prediction rates across gender groups - though this requires further investigation with more seeds in D4.

Recommended pipeline. Based on D3 results, we recommend **Prune (70%) → Quantize** as the practical compression pipeline for credit-scoring MLPs: it achieves the $4\times$ size reduction of quantization, the accuracy profile of pruning (0.750/0.815 on German Credit/UCI Adult), and the best or near-best fairness of all five pipelines on both datasets.

Limitations. (1) F1 instability at 70% sparsity (German Credit) persists in Prune→Quant, inherited from the pruning step. (2) The improved UCI Adult fairness under Prune→Quant is observed but not yet theoretically explained - D4 will investigate whether this holds across additional sparsity levels and seeds. (3) Quant→Prune’s no-op behavior is implementation-specific to PyTorch’s dynamic quantization API; other quantization frameworks may behave differently.

6 Conclusion

D3 establishes two contributions. First, we validate our D1/D2 results against four existing studies [1,2,4] with consistent, expected directional differences. Second, our improvement - a combined pruning+quantization order-effect study extending Han et al. [7] - shows that **order matters**: Prune→Quant preserves (German Credit) or improves (UCI Adult, by 26%) the fairness gains of pruning, while Quant→Prune is functionally equivalent to quantization alone due to architectural incompatibility between PyTorch’s pruning and quantization APIs. We recommend Prune→Quant as the practical pipeline for fair, compressed credit-scoring models, to be further validated in D4.

References

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Code Availability

All code, notebooks, and results for D1-D3 are available at: <https://github.com/zda24b023-create/Algorithmic-Fairness-in-Credit-Scoring>